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Seonggyu **Park**

Assistant Research **Scientist**

Curriculum Vita

I am primarily interested in developing hydrologic models to identify challenges in present and future water resources management. I am particularly interested in assessing spatial-temporal patterns of interactions between groundwater and surface water, improving water quality and sanitation in developing countries with severe water shortages, and developing new water conservation technologies regarding water catchment and harvesting systems. I am also a strong proponent of open and reproducible workflows.

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spark
Github
Twitter
Google Scholar

Education

2014 — 2018	Ph.D., in Civil Engineering, Groundwater Engineering. Colorado State University, Fort Collins, CO, United States *Dissertation:* Dissertation: Enhancement of Coupled Surface/Subsurface Flow Models in Watersheds: Analysis, Model Development, Optimization, and User Accessibility.
2004 — 2006	M.E. in Environmental Engineering. University of Suwon, Suwon, South Korea *Thesis:* A study on Adsorption Characteristics of VOCs on Activated Carbon.
1998 — 2004	B.E. in Environmental Engineering. University of Suwon, Suwon, South Korea

Brief professional history

Apr. 2021 —	**Assistant Research Scientist**, Texas A&M AgriLife Research, Temple, TX, United States
2018 — 2021	**Postdoctoral Research Associate**, Texas A&M AgriLife Research, Temple, TX, United States
2015 — 2018	**Graduate Research Assistant**, Ryan Bailey's Lab in Dept. of Civil Engineering, Colorado State University, Fort Collins, CO, United States
2011 — 2012	**IT & Computer Instructor**, Intercultural Institute of California (IIC), San Francisco, CA, United States
2006 — 2014	**Researcher**, HWASHIN Engineering & Bio Technology Co., Ltd., Incheon, South Korea

Research Experience

Icons link to additional content

Model Applications

- Assess Spatio-Temporal Patterns of Groundwater - Surface Water Interactions and Solute Transports in the Colorado River Basin using coupled APEX-MODLOW Model, (2020 — current). APEX-MODFLOW, PEST
- Assess Water, Salt, Fire and Land Mangement in the Colorado River Basin using APEX, (2020 — current). APEX
- Develop modelling frameworks for parameter estimation, sensitivity and uncertainty analysis on integrated hydrological models, (2020 — current). Python, PEST
- Develop SWAT-MODFLOW model to support resilient development and return on investment analyses in the Cubango-Okavango River Basin (CORB), (2020 — current). SWAT-MODFLOW, PEST
- Provided technical supports by developing database, framework, and analysis tools for the Real-Time Conservation Effects and Assessment Project (CEAP) - National Modelling provided by United States Department of Agriculture (USDA), (2018 — 2019). SWAT+, Python
- Assessed Surface Water and Groundwater Resources in the Middle Bosque, Texas Watershed using the coupled SWAT-MODFLOW model, (2017 — 2018). SWAT-MODFLOW, PEST
- Conducted research on spatial integration of SWAT and MODFLOW and optimizing the model with Spatio-temporal variation, (2017 — 2018). SWAT-MODFLOW, PEST
- Developed groundwater models with MODFLOW for several watersheds (Middle Bosque in Texas, United States, Missouri, Deer Creek in Missouri, United States, Neishaboor, Iran, and Limpopo, Southern Africa), (2015 — 2018). MODFLOW

Model Development

- Contributed to the development of a new integrated model to simulate hydrological processes in watershed systems, (2021). APEX-MODFLOW, Fortran
- Contributed to enhancing SWAT+ simulation of groundwater flow and groundwater-surface water interactions using MODFLOW routines, (2020). SWAT+MODFLOW, Fortran
- Contributed to enhancing SWAT-MODFLOW, (2017). SWAT-MODFLOW, Fortran

Software Development

- Develop a graphical user interface, a workflow implemented in QGIS, for application and evaluation of APEX-MODFLOW models, (2020—current). APEXMOD, Python, QT, QGIS
- Contribute to the development of APEX model optimization tool, (2020—current). APEX-CUT, Python, QT
- Developed a graphical user interface, a workflow implemented (as a plugin) in QGIS, for application and evaluation of SWAT-MODFLOW models, (2019). QSWATMOD, Python, QT, QGIS

Web applications

- Develop a web application for visualizing and analyzing Colorado River Basin (CRB) APEX-MODFLOW model performance. (2021 — current). Python
- Develop a web application for visualizing and analyzing Biomass in grazing lands. (2021 — current). Python
- Develop a web application for visualizing and analyzing APEX simulation results. (2021 — current). Python
- Developed a visualization tool for National SWAT+ model assessments. (2018 — 2019). SWAT+, Python

Peer-Reviewed Publications

2022

12. Zaremehrjardy, M., Victor, J., **Park, S.**, Smerdon, B., Alessi, D.S. and Faramarzi, M (2022). Assessment of snowmelt and groundwater-surface water dynamics in mountainous, foothill, and plain regions in northern latitudes, *Journal of Hydrology*. 127449. doi 10.1016/j.jhydrol.2022.127449

2021

11. Bailey, R.T., Tasdighi, A., **Park, S.**, Tavakoli-Kivi, S., Abitew, T., Jeong, J., Green, C.H. and Worqlul, A.W (2021). APEX-MODFLOW: A New Integrated Model to Simulate Hydrological Processes in Watershed Systems, *Environmental Modelling & Software*. *134*, 105093. doi 10.1016/j.envsoft.2021.105093

2020

10. Liu, W., **Park, S.**, Bailey, R.T., Molina-Navarro, E., Andersen, H.E., Thodsen, H., Nielsen, A., Jeppesen, E., Jensen, J.S., Jensen, J.B. and Trolle, D (2020). Quantifying the streamflow response to groundwater abstractions for irrigation or drinking water at catchment scale using SWAT and SWAT-MODFLOW, *Environmental Sciences Europe*. *32*, 1-25. doi 10.1186/s12302-020-00395-6

9. Liu, W., Bailey, R.T., Andersen, H.E., Jeppesen, E., Nielsen, A., Peng, K., Molina-Navarro, E., **Park, S.**, Thodsen, H. and Trolle, D (2020). Quantifying the effects of climate change on hydrological regime and stream biota in a groundwater-dominated catchment: A modelling approach combining SWAT-MODFLOW with flow-biota empirical models, *Science of The Total Environment*. *745*, 140933. doi 10.1016/j.scitotenv.2020.140933

8. Kim, S., Kim, S., Cho, J., **Park, S.**, Jarrín Perez, F.X. and Kiniry, J.R (2020). Simulated biomass, climate change impacts, and nitrogen management to achieve switchgrass biofuel production at diverse sites in US, *Agronomy*. *10*, 503. doi 10.3390/agronomy10040503

7. Bailey, R.T., **Park, S.**, Bieger, K., Arnold, J.G. and Allen, P.M (2020). Enhancing SWAT+ simulation of groundwater flow and groundwater-surface water interactions using MODFLOW routines, *Environmental Modelling & Software*. *126*, 104660. doi 10.1016/j.envsoft.2020.104660

6. Liu, W., Bailey, R.T., Andersen, H.E., Jeppesen, E., **Park, S.**, Thodsen, H., Nielsen, A., Molina-Navarro, E. and Trolle, D (2020). Assessing the impacts of groundwater abstractions on flow regime and stream biota: Combining SWAT-MODFLOW with flow-biota empirical models, *Science of the Total Environment*. *706*, 135702. doi 10.1016/j.scitotenv.2019.135702

2019

5. Mosase, E., Ahiablame, L., **Park, S.** and Bailey, R (2019). Modelling potential groundwater recharge in the Limpopo River Basin with SWAT-MODFLOW, *Groundwater for sustainable development*. *9*, 100260. doi 10.1016/j.gsd.2019.100260

4. Yen, H., **Park, S.**, Arnold, J.G., Srinivasan, R., Chawanda, C.J., Wang, R., Feng, Q., Wu, J., Miao, C., Bieger, K. and Daggupati, P (2019). IPEAT+: A built-in optimization and automatic calibration tool of SWAT+, *Water*. *11* (1681), doi 10.3390/w11081681

3. Liu, W., **Park, S.**, Bailey, R.T., Molina-Navarro, E., Andersen, H.E., Thodsen, H., Nielsen, A., Jeppesen, E., Jensen, J.S., Jensen, J.B. and Trolle, D (2019). Comparing SWAT with SWAT-MODFLOW hydrological simulations when assessing the impacts of groundwater abstractions for irrigation and drinking water, *Hydrology and Earth System Sciences*. *Discussions*, 1-51. doi 10.5194/hess-2019-232

2. **Park, S.**, Nielsen, A., Bailey, R.T., Trolle, D. and Bieger, K (2019). A QGIS-based graphical user interface for application and evaluation of SWAT-MODFLOW models, *Environmental modelling & software*. *111*, 493-497. doi 10.1016/j.envsoft.2018.10.017

1. Molina-Navarro, E., Bailey, R.T., Andersen, H.E., Thodsen, H., Nielsen, A., **Park, S.**, Jensen, J.S., Jensen, J.B. and Trolle, D (2019). Comparison of abstraction scenarios simulated by SWAT and SWAT-MODFLOW, *Hydrological Sciences Journal*. *64*, 434-454. doi 10.1080/02626667.2019.1590583

Manuscripts under review for publication

5. Saadatpour, A., Bailey, R., Izady, A., Amin, Ziaei, A. and **Park, S.** (under review). Quantifying the impact of climate change and irrigation management on groundwater in an arid region with intensive groundwater abstraction, *Hydrological Sciences Journal*.

4. **Park, S.**, Jaehak, J., Tracy Baker, Piotr Wolski, Tadesse Abitew, Mike Murray-Hudson, Sekgowa Motsumi, Colin Apse and Carlos Andrade (under review). Spatio-temporal variation of water resource and fluxes in the Okavango basin, *Water Resources Research*.

3. Bailey, R.T., Jeong, J., **Park, S.** and Green, C.H (under review). Simulating Salinity Transport in High-Desert Landscapes using APEX-MODFLOW-Salt, *Journal of Hydrology*.

2. Michael White, Jeff Arnold, Katrin Bieger, Peter Allen, Jungang Gao, Natalja Cerkasova, Mauro DiLuzio, Marilyn Gambone, **Park, S.**, David D. Bosch, Haw Yen (under review). Development of a Field Scale, SWAT+ Modeling Framework for the Contiguous US, *Journal of the American Water Resources Association*.

1. **Park, S.**, Jeong, J., Motter, E., and Bailey, R (under review). Introducing APEXMOD, A QGIS plugin for application and evaluation for the Enhanced APEX model, *Environmental modelling & software*.

Training Manuals

3. **Park, S.**, Bailey, R.T. and Bieger, K (2019). QGIS Interface for SWAT-MODFLOW (QSWATMOD), Blackland Research and Extension Center, Texas A&M AgriLife Research, Temple, TX

2. Bailey, R.T. and **Park, S.** (2019). SWAT-MODFLOW Tutorial Version 3 — Documentation for Preparing Model Simulations, Department of Civil and Environmental Engineering, Colorado State University: Fort Collins, CO, United States

1. **Park, S.** and Bailey, R.T (2017). SWAT-MODFLOW Tutorial—Documentation for Preparing Model Simulations, Department of Civil and Environmental Engineering, Colorado State University: Fort Collins, CO, United States

Technical Reports

2. **Park, S.**, Baker, T., Srinivasan, R. and Jeong, J (2022). SWAT-MODFLOW Relink Editor On Hydrologic And Water Quality System (HAWQS), BREC Report No. 22-01

1. **Park, S.**, Baker, T. and Jeong, J (2021). Development Of The Cubango-Okavango River Basin SWAT-MODFLOW Model, BREC Report No. 21-01

National & International Conference Presentations

2021

22. **Park, S.**, Jeong, J., Baker, B., Wolski, P., Abitew, T., Murray-Hudson, M., Motsumi, S., Apse, C., and Andrade, C (December, 2021). *Spatio-temporal variation of water resource and fluxes in the Okavango basin*, Poster presented at the AGU Fall Meeting 2021, New Orleans, LA, United States.

21. **Park, S.**, Abitew, TA., Jeong, J., Han, H., and Green, C (December, 2021). *Reproducible Parameter Estimation Framework for Integrated Hydrological Models as Evidenced in the Colorado River Basin*, Poster presented at the AGU Fall Meeting 2022, New Orleans, LA, United States.

20. H Han, TA Abitew, S Park, C Green, J Jeong (December, 2021). *A Machine Learning Approach for Assessing Wildfire Vulnerability in the Colorado River Basin*, Poster presented at the AGU Fall Meeting 2021, New Orleans, LA, United States.

2020

19. **Park, S.**, White, M.J., Gao, J., Bieger, K., and Arnold, J.G (August, 2020). *Analysis and visualization of large-scale model output with an interactive web-based dashboard*, at the 2020 Annual Water Resources Conference, Virtual.

2019

18. Sheng, Z., Ahn, S., Jung, C., **Park, S.**, Bailey, R., Granados-Olivas, A., Mirchi, A., Samimi, M. and Hargrove, W.L (December, 2019). *Coupled SWAT-MODFLOW Modeling for Determining Groundwater Sustainability Under Climate and Pumping Scenarios in a Semi-Arid Agricultural Watershed*, at the AGU Fall Meeting 2019, San Francisco, CA, United States.

17. **Park, S.**, Bailey, R.T., Schürz, C., Arnold, J.G., and Bieger, K (July, 2019). *Framework for quantifying uncertainty, sensitivity, and estimating parameters for SWAT+ with MODFLOW routines*, at the 2019 International SWAT Conference, Vienna, Austria.

16. Bailey, R.T., **Park, S.**, Navarro, E.M., Liu, W., Wei, X. and Trolle, D (July, 2019). *SWAT-MODFLOW: Recent Applications and an Introduction to Version 3*, at the 2019 International SWAT Conference, Vienna, Austria.

15. Bailey, R.T., **Park, S.**, Bieger, K., and Arnold, J.G (July, 2019). *Enhancing groundwater flow and groundwater-surface exchange simulation in SWAT+ using MODFLOW routines*, at the 2019 International SWAT Conference, Vienna, Austria.

14. Yen, H., **Park, S.**, Arnold, J.G., Chawanda, C.J. and Griensven, A.V (July, 2019). *IPEAT+: FORTRAN-based Automatic Calibration Tool Coupled with SWAT+*, at the 2019 International SWAT Conference, Vienna, Austria.

13. Chawanda, C.J., Yen, H., Arnold, J.G., Griensven, A.V., and **Park, S.** (July, 2019). *IPEAT+UI: An Automatic Calibration Tool with User Interface of SWAT+*, at the 2019 International SWAT Conference, Vienna, Austria.

2018

12. Ahn, S., Sheng, Z., **Park, S.**, and Bailey, R. T (December, 2018). *Surface Water-Groundwater modeling for Semi-Arid Agricultural Areas using SWAT-MODFLOW*, at the 2018 Fall Meeting AGU, Washington, D.C., United States.

11. **Park, S.**, Bailey, R.T., Bieger, K., and Arnold, J.G (September, 2018). *Coupling the SWAT+ and MODFLOW codes for enhanced surface / subsurface flow modeling in watersheds*, at the 2018 International SWAT Conference, Brussels, Belgium.

10. Molina-Navarro, E., Bailey, R.T., Andersen, H.E., Thodsen, H., Nielsen, A., **Park, S.**, Jensen, J. S., Jensen, J. B., and Trolle, D (September, 2018). *Comparison of performance and simulation of groundwater abstraction scenarios*, at the 2018 International SWAT Conference, Brussels, Belgium.

9. **Park, S.**, and Bailey, R. T (June, 2018). *Methodology for quantifying mode factor sensitivity, uncertainty, and estimation for integrated groundwater/surface water hydrologic models*, at the 9th International Congress on Environmental Modelling & Software, Fort Collins, CO, United States.

8. Bailey, R.T., **Park, S.**, Bieger, K., and Arnold, J.G (June, 2018). *A new model for simulating surface-subsurface hydrological processes at the watershed scale*, at the 9th International Congress on Environmental Modelling & Software, Fort Collins, CO, United States.

7. **Park, S.**, Nielsen, A., Bailey, R.T., Trolle, D., and Bieger, K (June, 2018). *A QGIS-based graphical user interface for application and evaluation of SWAT-MODFLOW models*, at the 9th International Congress on Environmental Modelling & Software, Fort Collins, CO, United States.

6. Saadatpour, A., Alizadeh, A., Ziaei, A.N., Bailey, R.T., **Park, S.**, Izady, A., and A.H. Zahraee (April, 2018). *A model for simulating coupled groundwater/surface water hydrology and impact of groundwater pumping in arid irrigated watersheds*, at the EGU General Assembly 2018, Vienna, Austria.

2017

5. Bieger, K., Bailey, R.T., **Park, S.**, and Arnold, J.G (November, 2017). *Development of an improved modeling framework for simulating nutrient transport in tile-drained watersheds*, at the 2017 AWRA Annual Conference, Portland, OR., United States.

4. Bailey, R.T, and **Park, S.** (Octobor, 2017). *SWAT-MODFLOW: Hydrologic and Water Quality Modeling Tool for Coupled Land Surface / Groundwater Systems*, at the ASA-CSSA-SSSA 2017, Tampa, FL, United States.

2016

3. Bailey, R.T, Wei, X., **Park, S.**, Rathjens, H., Bieger, K., Abbas, S., and Ajaaj, A (November, 2016). *Recent Advances in using SWAT-MODFLOW to Simulate Hydrological Processes in River Basins*, at the 2016 AWRA Annual Water Resources Conference, Orlando, FL, United States.

2. **Park, S.**, Bailey, R.T., and Bieger, K (July, 2016). *Quantifying Surface Water and Groundwater Resources in the Middle Bosque, Texas Watershed using the Coupled SWAT-MODFLOW Model*, at the 2016 International SWAT conference, Beijing, China.

1. Bailey, R.T., Rathjens, H., Bieger, K., **Park, S.**, and Wei, X (July, 2016). *SWATMOD-Prep: A Graphical User Interface for Preparing Coupled SWAT-MODFLOW simulations*, at the 2016 International SWAT conference, Beijing, China.

Funding Activity

Icons link to additional content

Under review

1. Development of a Novel Water Quality Index Considering Impacts of Natural and Anthropogenic Hazards using APEX-MODFLOW. January 2023 to December 2025. *Environmental Protection Agency (EPA)* . Proposed budget: $734,219. **Role:** Co-PI. PI: Jaehak Jeong. FTE: 0.00.

Current projects

1. Assessment of Water, Salt, Fire and Land Management in Colorado River Basin using APEX. May 2017 to April 2022. *Bureau of Land Management (BLM)* . Total budget: $1,004,036. **Role:** Assistant Research Scientist. PI: Jaehak Jeong. FTE: 1.00.

Professional Service

Workshops delivered

6. ArcAPEX for Agricultural Watershed Modeling Workshop | APEX-MODFLOW (3rd Day), number of attendees: 39, (Mar. 16-18, 2022). *NA*, Virtual.
5. Salinity Modeling Workshop (3 hours-session), number of attendees: 20, (Jan. 26, 2022). *NA*, Virtual.
4. SWAT-MODFLOW / QSWATMOD with Dr. Ryan Bailey, number of attendees: 11, (Jul. 15-16, 2019). *2019 International SWAT Conference*, Vienna, Austria.
3. SWAT-MODFLOW / QSWATMOD with Dr. Ryan Bailey, number of attendees: 20, (Sep. 17-18, 2018). *2018 International SWAT Conference*, Brussels, Belgium.
2. Introduction to QSWATMOD, number of attendees: 20, (Jun. 24, 2018). *9th International Congress on Environmental Modelling & Software*, Fort Collins, CO, United States.
1. Introduction to SWAT-MODFLOW / Model Optimization with PEST, number of attendees: 7, (Nov. 16, 2017). *Aarhus University*, Aarhus, Denmark.

Invited talks

1. Introduction to the SWAT-MODFLOW model, (Aug. 11, 2016). *Ulsan National Institute of Science and Technology (UNIST)*, Ulsan, South Korea.

Fellowships & Awards

3. Whitney Borland Scholarship, Colorado State University, $5000, (2017).
2. Whitney Borland Scholarship, Colorado State University, $5000, (2016).
1. Hwashin E&B Affiliated Research Institute Foundation Research Award, Hwashin E&B, $24000, (2014).

Professional Affiliations

2. American Geophysical Union
1. American Water Resources Association

Teaching

Icons link to additional content

Related experience

- Created lesson plans and trained people on software programs like Excel, Word, Access, Outlook, and PowerPoint by doing hands-on demonstrations and providing one-on-one application support.. (2011 — 2012).
- Provided undergraduate students with a broad introduction into 2-dimensional and 3-dimensional Computer-Aided Design (CAD) with a focus on construction and architecture-specific applications including environmental infrastructure. (2004 — 2006). *AutoCAD*
- Taught an industry-leading CAD software program (Autodesk AutoCAD) to model construction projects and then create and distribute basic, industry-standard architectural drawings. (2004 — 2006). *AutoCAD*